

RESEARCH PROPOSAL

A Controlled Benchmark of Temporal Graph Neural Networks for Gene Regulatory Network Inference Under Sparse Time-Series Conditions

Evaluating whether graph neural network methods for temporal GRN inference are more or less sensitive to limited time-series length than one another, under a unified evaluation protocol

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Abstract

Temporal graph neural networks (GNNs) are increasingly used to infer gene regulatory networks (GRNs) from time-series gene expression data, promising to capture the dynamic, cause-before-effect structure that static “snapshot” methods cannot. In the last two years, at least four such architectures — RiTINI, Marlene, PseudoGRN, and MTGRN — have each claimed state-of-the-art performance, but each was validated on its own choice of dataset, metrics, and baselines. No published study has compared them against one another under the same controlled conditions. This proposal outlines a systematic, head-to-head benchmark of these four methods, organized into two architectural groups (discrete snapshot models vs. continuous/pseudotime models), evaluated on shared synthetic (SERGIO-simulated) and real (time-series scRNA-seq) datasets. The core independent variable is temporal density: each method is evaluated as the number of available time points is progressively reduced, to determine which architectures degrade gracefully and which collapse under realistic, resource-constrained experimental conditions. The project requires no novel architecture and no manually constructed dataset — it contributes a rigorous, reproducible comparison that the field currently lacks, along with practical guidance for wet-lab researchers deciding how many time points a temporal-omics experiment actually needs.

1. Problem Definition

A gene regulatory network (GRN) describes which genes (specifically, transcription factors) control the activity of other genes. Most published GRN-inference methods treat this as a static problem: given one snapshot of gene expression, infer a single fixed wiring diagram. But real regulatory wiring is not fixed — it changes as a cell divides, differentiates, or responds to a stimulus. A static network is a single photograph; the underlying biology is a video. Knowing the temporal order in which genes activate (which regulator acts first, which responds later) is what allows researchers to identify early disease markers, guide stem-cell differentiation toward a target cell type, and time drug interventions correctly — questions a purely static network cannot answer.

To address this, a wave of temporal GNN architectures for GRN inference has emerged since 2023 (RiTINI, Marlene, PseudoGRN, MTGRN, among others), each combining graph-based structure learning with a mechanism for handling time (recurrent updates, neural ODEs, attention, or pseudotime). Each of these papers reports strong results — but each was tested in isolation: on its own dataset, its own baselines, and its own choice of metric. This is a well-documented pattern in computational biology benchmarking, where authors of a new architecture rarely compare fairly against the most recent competing architectures, only against older, weaker baselines.

The practical consequence is that a biologist, or a fellow researcher, currently has no reliable way to answer a simple question: “if I only have 3–5 time points of expression data (which is what most labs can realistically afford), which of these newer methods should I actually use?” That question is the problem this proposal addresses.

2. Research Gap and Motivation

2.1 What already exists

Two categories of prior work exist, and neither closes the gap:

- Static GRN benchmarks (e.g., BEELINE) compare classical and deep-learning methods on single-snapshot data, but do not include the newer temporal-GNN architectures and do not vary time-series length.

- Individual temporal-GNN papers (RiTINI, PseudoGRN, MTGRN, Marlene) each demonstrate that their own method handles sparse or dynamic data well, but every one of them reports results using its own full set of available time points — none tests what happens as that number shrinks, and none is compared directly against the others under identical conditions.

2.2 The specific, verified gap

This proposal targets a precise, checkable gap: “does GNN-based temporal GRN inference degrade differently across architectures as the number of available time points is reduced, and how does that degradation compare across the two dominant input paradigms (discrete snapshot vs. continuous/pseudotime)?” As of the literature check underlying this proposal (mid-2026), no publicly indexed paper performs this specific four-way, temporal-density-controlled comparison. This should be re-verified with a fresh literature search immediately before manuscript writing begins, since this sub-field is active and moves quickly.

2.3 Why this matters beyond the methods literature

- Disease progression: ordering of gene activation identifies which genes are causal early markers versus downstream effects.
- Regenerative medicine: knowing the correct activation order helps guide stem cells toward a target lineage.
- Experimental design / cost: sequencing additional time points is expensive; if a method needs very few time points to remain accurate, that is directly actionable guidance for a wet lab.

3. Research Objectives

The overarching goal of this project is to produce the first controlled, reproducible, head-to-head benchmark of modern temporal-GNN methods for GRN inference, focused specifically on robustness to sparse time-series data. Concretely, the objectives are:

- O1 — Assemble a unified evaluation pipeline that can feed the same underlying dataset, in the two formats each architecture expects (discrete snapshot vs. continuous/pseudotime), to all four candidate models.
- O2 — Quantify each model's structural accuracy (edge prediction quality) on both synthetic (ground-truth-known) and real (silver-standard) datasets.
- O3 — Systematically vary the number of available time points (temporal density) and measure each model's performance decay curve, identifying break-points where accuracy collapses.
- O4 — Compare degradation patterns across the two architectural groups (snapshot-based vs. pseudotime-based) to determine whether one input paradigm is inherently more data-efficient.
- O5 — Evaluate computational cost (training time, memory) alongside accuracy, since a method that needs disproportionate compute for marginal accuracy gains is less useful in practice.
- O6 — Translate findings into practical guidance (e.g., “method X remains reliable down to 3 time points; method Y requires at least 6”) usable by experimentalists planning a temporal-omics study.

4. Research Questions

RQ1 (Primary):

How does the GRN-inference accuracy of four representative temporal-GNN architectures (RiTINI, Marlene, PseudoGRN, MTGRN) change as the number of available time points is reduced from dense (≥ 6) to ultra-sparse (2–3)?

RQ2:

Do discrete snapshot-based models (Group A) and continuous/pseudotime-based models (Group B) degrade differently as data becomes sparse, and if so, which paradigm is more robust?

RQ3:

Is there a consistent “break-point” (minimum number of time points) below which all four methods fail to outperform a simple classical baseline (e.g., GENIE3), and does this break-point differ between synthetic and real data?

RQ4:

Does higher structural accuracy come at a proportionally higher computational cost, and which method offers the best accuracy-per-compute trade-off under the Kaggle-scale free-GPU budget this project operates within?

Working hypothesis:

Based on general trends in other domains (GNNs typically require more training signal than classical statistical methods), we hypothesize that at least some GNN-based methods will underperform classical baselines under ultra-sparse conditions — a negative result would itself be a valid and useful contribution, not a failure of the project.

5. Proposed Methodology

5.1 Overall design

The project is a controlled comparative benchmark, not a new-architecture paper. The methodology has four pillars: (i) a two-group model taxonomy that respects each method's native input format, (ii) a dual data strategy (synthetic + real), (iii) a single controlled variable — temporal density — applied identically across all methods, and (iv) a uniform pre-processing, tuning, and evaluation protocol so that no method is unfairly advantaged.

5.2 Model selection and grouping

All four candidate methods target the same underlying biological problem — inferring directed regulator → target-gene interactions that change over time — but split into two groups based on the input format they expect. This split must be respected in the benchmark design; the two groups are not thrown into one undifferentiated pool.

Model	Group	Core mechanism	Published
RiTINI	A — Snapshot	Graph attention + Graph Neural ODE; continuous-time interpolation between sparse intervals	2023–2024
Marlene	A — Snapshot	Self-attention with GRU-evolved weights; meta-learning for rare cell types	2024–2025
PseudoGRN	B — Continuous	Unifies pseudotime inference methods with time-lagged divergence and partial correlation	2025

MTGRN	B — Continuous	Transformer-based; frames GRN inference as multivariate forecasting along a lineage trajectory	2024
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Group A (snapshot models) consumes discrete, aggregated expression matrices at labeled macro time points (e.g., Day 0, Day 2, Day 5). Group B (continuous/pseudotime models) consumes raw single-cell data with real timestamps and computes its own internal pseudotime ordering. Both groups are answering the same biological question with the same class of underlying data — only the pre-processing step differs — which keeps the comparison valid while respecting each architecture's design.

5.2.1 Screening check: why DA-RNN, GRGNN, and DGCGRN were excluded

Because AUPRC/AUROC scoring requires every candidate to output the same kind of object — a directed edge-probability matrix, produced from time-indexed input and scored against a ground-truth network — every candidate model considered for this benchmark was screened against two questions before being admitted to the final list: (1) does the architecture natively consume a time-indexed sequence or pseudotime trajectory, rather than a single static snapshot; and (2) does it natively produce and get evaluated on a graph structure, rather than requiring a graph to be extracted after the fact from parameters trained for a different objective? A model failing either question is not a like-for-like comparison against the others and was removed rather than kept with a caveat.

DA-RNN (Dual-Stage Attention RNN) fails question 2. It is a time-series forecasting model — trained to predict a target gene's future expression value, not to predict edges. The only way to obtain a network from it is to repurpose its input-attention weights as an implied adjacency matrix, a workaround rather than a designed output. This is not a hypothetical concern: the paper that pioneered this exact workaround for GRNs (Randazzo et al., Entropy, 2022) reports its own negative result — the input attention mechanism of the DA-RNN does not reflect the causal interactions between the variables composing the system, and the authors state that looking at the internal parameters of the attention layer, they could not fully reconstruct the internal connection among the genes. They used the attention matrix only to compute coarse graph-theoretic descriptors (clustering coefficient, hub score) to distinguish network archetypes — never to score precision/recall against a known wiring diagram, which is exactly what this benchmark requires.

This is a documented failure mode beyond DA-RNN specifically. A 2026 interpretability study evaluating attention-derived GRN predictions from large single-cell foundation models (scGPT, Geneformer) found they scored at or near chance against real reference networks — AUROC values ranged from 0.444 to 0.549 against TRRUST, with bootstrap confidence intervals that included 0.50 (i.e., statistically indistinguishable from a coin flip) — confirming that attention weight is not a reliable stand-in for a trained, supervised edge signal in this problem domain generally, not only in the DA-RNN case.

GRGNN and DGCGRN both fail question 1. GRGNN is not a temporal model at all: it formulates GRN inference as a one-shot graph classification problem, deciding whether a subgraph centered on a candidate TF–target pair contains a true edge. Its own paper states this requires a noisy starting graph structure built from partial information, such as Pearson's correlation coefficient and mutual information — meaning it needs a pre-built candidate network as input, not a time-indexed expression sequence, with gene expression entering only as a static per-node feature vector across conditions. GRGNN is also confirmed undirected-only: as this study aims to predict link existence, E is always treated as an undirected edge in our formulation. DGCGRN was initially proposed as a replacement for DA-RNN on the reasoning that its Bi-GRU “sequence features” handled temporal dynamics; on closer reading this was an error in an earlier draft of this proposal — those sequence features are extracted from each gene's own nucleotide sequence and its physicochemical characteristics, not from a time series of expression measurements. DGCGRN builds directly on GRGNN's graph-convolutional, candidate-skeleton approach, extending it to directed edges, and inherits the same static, single-snapshot

formulation. Both are legitimate, well-evaluated methods for their actual task — static/semi-supervised GRN inference — but that is a different task from the one this benchmark tests. Including them would silently broaden the research question from “how do temporal-GNN methods handle sparse time series” to “how do temporal and static GNN methods compare,” which was not the goal and would require its own, separately justified selection of static-model representatives (e.g., DeepFGRN, XATGRN, or classical GENIE3) rather than whichever two happened to already be on this list.

The remaining four candidates were checked against both questions and confirmed valid on both: RiTINI's attention is trained under an explicit graph regularization loss over a sequence of interpolated time steps and evaluated against a known 100-gene, 137-edge network; Marlene's self-attention output is explicitly described as generating a gene network in the form of an adjacency matrix from time-varying input; PseudoGRN reports AUROC and AUPRC to assess the accuracy of predicted regulatory edges directly, inferred from pseudotime-ordered data; and MTGRN, despite using a forecasting mechanism internally, routes that forecast through a dedicated structural module operating over a cell-lineage trajectory and reports a directly comparable AUPRC of 0.639 on real ground-truth data. Scoping the benchmark to these four keeps the original research question — temporal robustness under data scarcity — intact, at the cost of a smaller model count, which is treated as an explicit, stated limitation (Section 7) rather than a hidden one.

5.3 Selection justification (anticipated reviewer question: “why these four?”)

- **Shared problem definition:** all four explicitly target temporal / dynamic GRN inference from time-indexed or pseudotime-ordered data (directed TF → target-gene edges that change over time), not static, single-snapshot inference.
- **Shared architecture family:** all four are graph-neural or attention-based deep learning methods published 2023–2025, explicitly excluding older non-deep-learning baselines (SCODE, GENIE3, LEAP), which are retained only as reference baselines, not benchmark subjects.
- **Explicit data-scarcity framing:** each of the four explicitly motivates itself around limited temporal resolution or data-starved conditions in its own abstract — making “how well do you actually handle scarcity” a fair question to ask of all of them.
- **Open-source availability:** all four have public code or are reproducible from their published architecture description, which is a hard requirement — methods without accessible implementations are excluded regardless of relevance.
- **Static/semi-supervised methods excluded by design:** GRGNN, DGCGRN, and similar candidate-skeleton graph classifiers (e.g., DeepFGRN, XATGRN) are out of scope because they do not consume time-indexed input, which is the axis this benchmark tests — a scoping decision documented in Section 5.2.1, not an oversight.
- **Excluded by design:** methods requiring modalities beyond temporal transcriptomics (e.g., STAGED, which needs spatial transcriptomics; Dictys, which needs paired scATAC-seq) are explicitly excluded and this exclusion is stated as a scoping decision in the paper, not an oversight.

5.4 Dataset strategy

No dataset needs to be manually constructed. Two complementary, fully public data sources are used:

Type	Source	Role in the benchmark
Synthetic (ground truth known)	SERGIO or GeneNetWeaver simulators; parameterized to generate expression time series with a known true network	Primary quantitative evaluation — exact precision/recall against a flawless answer key; used to build the main temporal-density degradation curves

Real biological (high complexity)	Public time-series scRNA-seq, e.g. mouse embryonic stem cell (mESC) differentiation (Day 0/2/4/7) or human hematopoiesis	External validity check; predictions scored against silver-standard reference edges from TRRUST / ChIP-Atlas / STRING
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For Group A, the same real dataset is aggregated into per-day mean-expression matrices with a short pre-processing script. For Group B, the same raw cell-by-gene matrix with real day labels is passed through unmodified, and each method computes its own trajectory ordering. This ensures both groups see the same underlying biology, differing only in the pre-processing their architecture requires.

5.5 Controlled variable: temporal density

The single independent variable manipulated across all experiments is the number of time points supplied to each model. Time points are removed by slicing the existing dense series — no new data is collected.

Tier	Time points	Purpose
Tier 1 — Dense	6–21 (full series)	Baseline / control condition, matching what each original paper reported
Tier 2 — Sparse	4–5	Approximates a realistic, moderately funded lab experiment
Tier 3 — Ultra-sparse	2–3	Stress test; identifies the break-point where methods fail

As a secondary, lower-priority check, uneven sampling intervals (e.g., Day 0, Day 1, Day 7) are also tested against evenly spaced intervals, since real experiments rarely sample perfectly evenly.

5.6 Fair-comparison protocol

Because reviewers of benchmark papers scrutinize fairness above all else, the following is treated as non-negotiable methodology, not an implementation detail:

- Identical inputs: a single wrapper pipeline feeds every model the same underlying matrices (in the format each group requires), rather than re-implementing each model's own data loader.
- Identical hyperparameter search budget: every model receives the same number of automated tuning trials (e.g., 50 trials via Optuna) — no model is hand-tuned while another runs on defaults.
- Identical cross-validation scheme: a leave-one-time-point-out cross-validation scheme is applied uniformly.
- Isolated environments: each model runs in its own containerized/virtual environment to avoid dependency conflicts between codebases from different labs, without needing to rewrite any model's internal code.
- Neutral reporting language: differences are reported strictly as quantitative deltas (e.g., “a 42% drop in AUPRC when time points are reduced from 5 to 3”), never as qualitative judgments about a model being “bad.”

5.7 Evaluation metrics

- Structural accuracy: Area Under the Precision-Recall Curve (AUPRC) as the primary metric (standard for GRN inference, since true networks are sparse), plus AUROC and top-K edge accuracy as secondary metrics.
- Topology preservation: whether predicted networks retain known biological structure, e.g., scale-free degree distribution and correct identification of hub transcription factors.
- Computational efficiency: wall-clock training time and peak memory usage as the number of genes and time points scale.
- Degradation profile: accuracy plotted against temporal-density tier per model, per group — the central figure of the paper.

6. Feasibility Assessment

6.1 Literature / novelty feasibility

A comprehensive benchmark comparing these four specific temporal-GNN architectures against each other under systematic temporal degradation was not found in the literature as of the check underlying this proposal. Existing related work either benchmarks static or classical dynamic methods (e.g., BEELINE-style comparisons against SCODE, DynGENIE3) or is a single-architecture paper compared only to older, non-competing baselines. This is a genuinely open, verifiable niche — but it is worth flagging that this exact sub-field is active (at least one directly adjacent paper, on label-scarcity meta-learning for GRNs, was published within months of this proposal being drafted), so the literature check should be repeated immediately before manuscript submission, not only at the planning stage.

6.2 Computational feasibility (Kaggle free GPU)

Kaggle provides roughly 30 free weekly hours on dual NVIDIA T4 GPUs (16GB VRAM each), which is sufficient for this project provided the problem domain — not the models — is downscaled:

- Lighter models (Marlene, PseudoGRN): attention/recurrent-based over sparse matrices; expected to train within minutes to a few hours per run on a T4.
- Heavier models (RiTINI, MTGRN): ODE solvers and transformer blocks are more memory- and time-intensive; mitigated by restricting all experiments to the top 500–1,000 highly variable genes or known transcription factors rather than the full ~20,000-gene transcriptome. Benchmarking at this gene-set scale is standard practice in the computational-biology literature, not a compromise on rigor.
- If a specific run still exceeds Kaggle's session limits, Google Colab's free tier or a university compute allocation (if available) serves as a fallback for that run only, without changing the experimental design.

6.3 Personal feasibility as a graduate researcher

Writing a benchmark paper is an achievable and strategically sound choice for an early-career researcher: it does not require inventing a new architecture (lower technical risk), and it forces deep, hands-on understanding of four recent architectures through reimplementation and reproduction, which has direct pedagogical value. The main non-technical risk is defensibility — reviewers (who may include an original author of one of the four methods) will scrutinize any perceived unfairness. This is mitigated directly by the fair-comparison protocol in Section 5.6: neutral, quantitative language; identical tuning budgets; and a dedicated fairness section in the eventual manuscript.

6.4 Outcome risk is low by design

Because this is a benchmarking study rather than a novel-method study, there is no scenario in which the project “fails” to produce a publishable result. If one method dominates, that is a result. If all four methods collapse below a certain number of time points, that is arguably a more important finding — it would indicate the whole sub-field is over-fit to the dense-time-point conditions in which it was originally validated, which is itself a legitimate and citable conclusion.

7. Risks and Mitigation

Risk	Likelihood / Impact	Mitigation
A directly overlapping benchmark is published by another group before submission	Moderate / High — sub-field is active	Re-run the literature check monthly; prioritize early, incremental result-sharing (e.g., a workshop paper or preprint) over a single late submission
A model's public code is broken, undocumented, or unmaintained	Moderate / Medium	Verify all four repositories run end-to-end on toy data before committing further time; keep a shortlist of 1–2 backup models per group identified in advance
Four-model sample size is seen as too small by reviewers	Low–Moderate / Medium	State explicitly in the paper that scope was narrowed deliberately from six to four candidates after two (GRGNN, DGCGRN) failed a pre-registered temporal-input screening check (Section 5.2.1) — framed as a rigor decision, not a shortfall; results remain a full 2×2 factorial across both input paradigms
Kaggle GPU time is insufficient for heavier models (RiTINI, MTGRN)	Moderate / Medium	Restrict to top 500–1,000 genes; use Colab as overflow; consider reducing Optuna trial budget uniformly (still fair, since applied to all models equally) if needed
Reviewers perceive the comparison as unfair to one method's original authors	Low–Moderate / High	Strict adherence to the fair-comparison protocol (Section 5.6); pre-register the evaluation protocol before running final experiments
Real-data ground truth (silver-standard networks) is noisy or incomplete	High / Medium	Treat synthetic (SERGIO) results as the primary quantitative evidence; report real-data results as a supporting, more loosely validated check, explicitly labeled as such

8. Proposed Timeline

An illustrative timeline assuming a part-time graduate research schedule; adjust to your actual semester/term structure.

Phase	Duration	Activities
Phase 1	Weeks 1–2	Repeat literature check; download and sanity-test all four model repositories on toy/tutorial data; confirm each runs end-to-end
Phase 2	Weeks 3–4	Acquire and pre-process datasets: SERGIO simulation setup; mESC (or equivalent) real dataset download; silver-standard reference network preparation
Phase 3	Weeks 5–6	Build the unified wrapper pipeline (shared input format, Optuna tuning harness, evaluation/metric code)
Phase 4	Weeks 7–10	Run full experimental grid: 4 models $\times$ 2–3 datasets $\times$ 3 temporal-density tiers; log accuracy, topology, and compute metrics
Phase 5	Weeks 11–12	Analysis: degradation curves, Group A vs. Group B comparison, break-point identification, statistical significance checks
Phase 6	Weeks 13–14	Draft manuscript; supervisor review cycle; final literature re-check before submission

9. Expected Contributions

- The first controlled, reproducible, four-way benchmark of modern temporal-GNN GRN-inference methods, evaluated under identical conditions rather than each method's self-reported setup.
- A quantified answer to a practically important question for experimentalists: how few time points can a temporal-omics experiment collect before these methods stop being trustworthy?
- A direct comparison between the two dominant input paradigms (discrete snapshot vs. continuous/pseudotime) for data efficiency, which no existing paper isolates.
- A reusable, open evaluation pipeline (unified pre-processing, tuning, and metrics) that future GRN-inference papers can adopt instead of each inventing their own comparison protocol.
- Either a clear ranking of which architecture is most robust to data scarcity, or — equally valuable — evidence that the entire current generation of temporal-GNN methods is over-fit to dense time-series conditions.

10. Key References

(Representative sources identified during problem formulation; a full systematic literature review should be conducted and this list expanded before manuscript submission.)

- Bhaskar, D. et al. “Inferring dynamic regulatory interaction graphs from time series data with perturbations” (RiTINI). arXiv:2306.07803 / PMLR.

- “Recovering time-varying networks from single-cell data” (Marlene). PMC12261490 / arXiv:2410.01853.
- Wang, J. et al. “Inductive Inference of Gene Regulatory Network Using Supervised and Semi-supervised Graph Neural Networks” (GRGNN; cited to justify its exclusion as a static, non-temporal method, not as a benchmark subject). bioRxiv 2020.09.27.315382.
- Wei, T. et al. “Inference of gene regulatory networks based on directed graph convolutional networks” (DGCGRN; cited to justify its exclusion as a static, non-temporal method, not as a benchmark subject). Briefings in Bioinformatics, 2024. PMC11209731.
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- “Systematic Evaluation of Single-Cell Foundation Model Interpretability Reveals Attention Captures Co-Expression Rather Than Unique Regulatory Signal.” 2026 (arXiv:2602.17532) — supporting evidence that attention-derived edges are an unreliable substitute for trained link-prediction output.
- “Gene Regulatory Network Inference from Pseudotime-Ordered scRNA-seq Data via Time-Lagged Divergence Measures” (PseudoGRN). ICBRA proceedings, 2025 / PMC12756061.
- “Deciphering Cell Lineage Gene Regulatory Network via MTGRN.” OpenReview, 2024.
- “Structure-enhanced graph meta learning for few-shot gene regulatory network inference” (Meta-TGLink; related but distinct scarcity axis — label scarcity, not temporal scarcity). Genome Biology, 2025.
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- Dibaeinia, P. & Sinha, S. “SERGIO: a single-cell expression simulator guided by gene regulatory networks.” Cell Systems, 2020.
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